

# Trevor J. David

---

## CONTACT

☎ +1 (914) 393 3293  
✉ [trevorjdavid@gmail.com](mailto:trevorjdavid@gmail.com)  
🐙 [github.com/trevordavid](https://github.com/trevordavid)  
🌐 [linkedin.com/in/trevorjdavid](https://linkedin.com/in/trevorjdavid)

## EDUCATION

2018 PH.D. IN ASTROPHYSICS, CALIFORNIA INSTITUTE OF TECHNOLOGY  
Thesis: *On the Evolutionary Pathways of Stars and Extrasolar Planets*  
2013 M.S. IN ASTROPHYSICS, CALIFORNIA INSTITUTE OF TECHNOLOGY  
GPA: 3.8/4.0  
2009 B.A. IN PHYSICS & ASTRONOMY, VASSAR COLLEGE  
GPA: 3.93/4.0

## APPOINTMENTS & EXPERIENCE

|              |                                                        |                                                                                |
|--------------|--------------------------------------------------------|--------------------------------------------------------------------------------|
| 2025–present | Vice President, DeFi Quantitative Research             | Galaxy                                                                         |
| 2022–2024    | Quantitative Researcher & Strategy Owner               | Linden Shore                                                                   |
| 2019–2022    | Flatiron Research Fellow                               | Center for Computational Astrophysics<br>Flatiron Institute, Simons Foundation |
| 2019–2022    | Research Associate                                     | American Museum of Natural History                                             |
| 2017–2019    | Exoplanetary Science Initiative<br>Postdoctoral Fellow | NASA Jet Propulsion Laboratory                                                 |
| 2017         | Graduate Research Assistant                            | California Institute of Technology                                             |
| 2014–2017    | NSF Graduate Research Fellow                           | California Institute of Technology                                             |
| 2011–2013    | Jesse L. Greenstein Fellow                             | California Institute of Technology                                             |
| 2009–2011    | Senior Editor and Science Writer                       | Atlas Obscura                                                                  |

## RESEARCH SKILLS

**Languages:** Python (14 years), C++ (2 years), SQL; **AI/ML:** scikit-learn, jax, PyMC, PyTorch; **Data manipulation & analysis:** pandas, polars; **Regression & classification:** gradient descent, linear regression (lasso, ridge), logistic regression, decision trees (Random Forest, LGBM), SVMs, MLPs, LLMs; **Probabilistic inference:** Markov chain Monte Carlo, Hamiltonian Monte Carlo, Gaussian process regression, Bayesian hierarchical modeling, Nested sampling; **Time series analysis:** signal processing, smoothing, detrending, filtering; **Software development:** version control (git), shell scripting (bash, zsh), scheduled tasks (cron), parallelization (joblib, multiprocessing, dask), containerization (docker), CI/CD (GitHub Actions).

## RESEARCH EXPERTISE

Formation and evolution of planetary systems; exoplanet demographics; transiting exoplanets; stellar evolution; atmospheric physics; open clusters and young stellar associations; fundamental stellar parameters; stellar ages; eclipsing binaries; time series analysis of photometric and radial velocity data.

## AWARDS & HONORS

|      |                                                                                                                                                            |                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2019 | Best Postdoc Research Poster (Astrophysics & Space Science)<br>– <i>Awarded to best research poster at NASA JPL Postdoc Research Day.</i>                  | NASA JPL       |
| 2016 | David & Barbara Groce Research Award<br>– <i>Scholarship awarded to support original research in astrophysics.</i>                                         | Caltech        |
| 2015 | Gerald Neugebauer Scholarship<br>– <i>Awarded to 1–3 graduate students annually to support research-related expenses.</i>                                  | Caltech        |
| 2011 | National Science Foundation Graduate Research Fellowship<br>– <i>Awarded to top 15% of graduate students nationally to support basic research in STEM.</i> | NSF            |
| 2009 | Lucy Kellogg English Prize for Excellence in Physics<br>– <i>Awarded to the top graduating physics major.</i>                                              | Vassar College |
| 2009 | Phi Beta Kappa<br>– <i>Awarded to distinguished students in the top 10% of their graduating class.</i>                                                     | Vassar College |
| 2009 | Sigma Xi<br>– <i>Awarded to graduating seniors based on research accomplishments and academic record.</i>                                                  | Vassar College |
| 2009 | Departmental Honors, Physics & Astronomy<br>– <i>Awarded to graduating seniors with excellent academic and research achievements.</i>                      | Vassar College |
| 2009 | General Honors<br>– <i>Awarded to students in the top 20% of their graduating class.</i>                                                                   | Vassar College |
| 2009 | Vera Rubin Research Award<br>– <i>Scholarship awarded to support original research in astrophysics.</i>                                                    | Vassar College |
| 2008 | Ethel Hickox Pollard Memorial Physics Award<br>– <i>Awarded to the junior physics major with the highest academic average.</i>                             | Vassar College |

## GRANTS & FUNDING AWARDS

| DATE | SOURCE       | AMOUNT    | PROJECT                                                               |
|------|--------------|-----------|-----------------------------------------------------------------------|
| 2019 | NASA TESS GI | \$100,000 | The Evolution of Dust and Debris Around Sun-like Stars (PI: David)    |
| 2018 | NASA K2 GO   | \$100,000 | Investigating the Early Evolution of Exoplanet Properties (PI: David) |
| 2012 | NSF GRFP     | \$132,000 | Age Determination of Prospective Exoplanet Host Stars (PI: David)     |

## TELESCOPE AWARDS

| DATE      | FACILITY         | TIME        | PROJECT                                                             |
|-----------|------------------|-------------|---------------------------------------------------------------------|
| 2021      | <i>JWST</i>      | 25 hours    | The nature, origin, and fate of two planets of a newborn system     |
| 2021      | <i>TESS</i>      | 2 sectors   | *Weighing newborn planets in the V1298 Tau System                   |
| 2020      | <i>Hubble</i>    | 8 orbits    | Atmospheric characterization of infant multiplanet system V1298 Tau |
| 2020      | <i>Hubble</i>    | 17 orbits   | Planetary mass loss and the high-energy spectrum of V1298 Tau       |
| 2020      | <i>Hubble</i>    | 10 orbits   | Water and methane in a juvenile transiting exoplanet                |
| 2019B     | Keck             | 1 night     | *Exoplanet transmission spectroscopy                                |
| 2019B     | Keck             | 1 night     | Spin-orbit alignment of a young exoplanet                           |
| 2019      | Palomar          | 13 nights   | PARVI Commissioning and early science                               |
| 2019      | <i>Spitzer</i>   | 12 hours    | *Characterizing the atmosphere of a newborn Jupiter-sized exoplanet |
| 2019A     | Subaru           | 2 nights    | Precision radial velocities of young stars                          |
| 2019A     | Keck             | 3 nights    | *Exoplanet transmission spectroscopy                                |
| 2018-2019 | Palomar          | 3 nights    | *Time series polarimetry of young low-mass stars                    |
| 2018      | <i>Kepler/K2</i> | Campaign 18 | *Planet occurrence in a young stellar cluster                       |
| 2018B     | Keck             | 1 night     | *Exoplanet transmission spectroscopy                                |
| 2018B     | Keck             | 1 night     | Spin-orbit alignment of a young exoplanet                           |
| 2017      | <i>Kepler/K2</i> | Campaign 15 | Disks, activity, and planets in Sco-Cen                             |
| 2017      | <i>Hubble</i>    | 10 orbits   | K2-33b transmission spectroscopy                                    |
| 2017      | <i>Spitzer</i>   | 23 hours    | Rapidly rotating young stars                                        |
| 2017A     | Keck             | 2 nights    | Spin-orbit alignment of a young binary                              |
| 2016-2017 | Keck             | 10.5 nights | *Pre-main-sequence binaries                                         |
| 2016      | <i>Kepler/K2</i> | Campaign 13 | Monitoring young stars in Taurus                                    |
| 2015-2016 | LCO              | 55 hours    | *Transiting debris orbiting young stars                             |
| 2013A     | Keck             | 1 night     | Exoplanet transmission spectroscopy                                 |
| 2012-2013 | Palomar          | 34 nights   | Directly imaging exoplanets                                         |
| 2012-2013 | Palomar          | 7 nights    | Stellar characterization                                            |
| 2009      | Apache Point     | 1 night     | Galactic archaeology                                                |
|           |                  |             | *As P.I. or Science P.I.                                            |

## COLLOQUIA & INVITED TALKS

| DATE    | EVENT                                              | VENUE                                          |
|---------|----------------------------------------------------|------------------------------------------------|
| 08/2022 | NASA Small Explorer Mission Concept Workshop       | NASA JPL, Pasadena, CA                         |
| 07/2021 | Sagan Exoplanet Summer Workshop                    | NASA Exoplanet Science Institute, Pasadena, CA |
| 02/2021 | Center for Exoplanets and Habitable Worlds Seminar | Penn State Univ., State College, PA            |
| 04/2020 | Exoplanet Meeting (Winn Group)                     | Princeton Univ., Princeton, NJ                 |
| 02/2020 | Astrophysics Seminar                               | AMNH, New York, NY                             |
| 11/2019 | ITC Stars & Planets Seminar                        | Harvard-Smithsonian CfA, Cambridge, MA         |
| 03/2019 | Exoplanetary Science Initiative Symposium          | Caltech, Pasadena, CA                          |
| 03/2019 | Astrophysics Seminar                               | Carnegie Observatories, Pasadena, CA           |
| 10/2018 | Astronomy Department Colloquium                    | Université de Montréal, QC                     |
| 09/2018 | ExSoCal 2018                                       | Caltech, Pasadena, CA                          |
| 08/2018 | Exoplanets Seminar                                 | NASA JPL, Pasadena, CA                         |
| 05/2018 | Planetary Science Seminar (Yung Group)             | Caltech, Pasadena, CA                          |
| 03/2018 | Exoplanetary Science Initiative Symposium          | NASA JPL, Pasadena, CA                         |
| 01/2018 | K2 Dwarf Stars and Clusters Workshop               | Boston University, Boston, MA                  |
| 09/2016 | Center for Integrative Planetary Science Seminar   | Univ. of California, Berkeley, CA              |
| 04/2016 | K2 CHAI Collaboration Meeting                      | Univ. of Hawai'i at Mānoa, Honolulu, HI        |

## CONTRIBUTED & OTHER TALKS

| DATE    | EVENT                              | VENUE                                                   |
|---------|------------------------------------|---------------------------------------------------------|
| 04/2022 | CCA Weekly Seminar                 | Flatiron Institute, New York, NY                        |
| 03/2022 | Fifty Years of Skumanich Relations | Boulder, CO                                             |
| 04/2021 | CCA Weekly Seminar                 | Flatiron Institute, New York, NY                        |
| 02/2020 | CCA Weekly Seminar                 | Flatiron Institute, New York, NY                        |
| 08/2019 | Extreme Solar Systems IV           | Harpa, Reykjavík, Iceland                               |
| 04/2018 | STScI Spring Symposium             | Space Telescope Science Institute, Baltimore, MD        |
| 02/2017 | K2 CHAI Collaboration Meeting      | NASA Exoplanet Science Institute, Pasadena, CA          |
| 01/2017 | American Astronomical Society 229  | Gaylord Texan Resort & Convention Center, Grapevine, TX |
| 10/2016 | DPS 48/EPSC 11                     | Pasadena Convention Center, Pasadena, CA                |
| 09/2016 | ExSoCal                            | Caltech, Pasadena, CA                                   |
| 09/2016 | Keck Science Meeting               | Caltech, Pasadena, CA                                   |
| 06/2016 | Cool Stars XIX                     | Uppsala Konsert & Kongress, Uppsala, Sweden             |
| 02/2016 | Exoplanet Meeting (Mawet Group)    | Caltech, Pasadena, CA                                   |
| 05/2015 | Exoplanet Meeting (Mawet Group)    | Caltech, Pasadena, CA                                   |

## TEACHING EXPERIENCE

| DATE | INSTITUTION                   | ROLE               | COURSE                                 |
|------|-------------------------------|--------------------|----------------------------------------|
| 2014 | Franklin Educational Services | Tutor              | Physics and Mathematics                |
| 2013 | Caltech                       | Teaching Assistant | Ay. 1: The Evolving Universe           |
| 2013 | Caltech                       | Teaching Assistant | Ay. 21: Galaxies and Cosmology         |
| 2012 | Caltech                       | Teaching Assistant | Ay. 20: Basic Astronomy and the Galaxy |
| 2009 | Poughkeepsie High School      | Teaching Assistant | Introductory Physics                   |
| 2009 | Vassar College                | Teaching Assistant | Phys. 280: Contemporary Optics         |
| 2008 | Vassar College                | Teaching Assistant | Phys. 180: Science of Sound            |
| 2008 | Vassar College                | Tutor              | Introductory Physics                   |

## ADVISING & MENTORSHIP

|      |                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | John Livingston (National Astronomical Observatory of Japan)<br>Project: <i>Constraining Planetary Evolution Models with Young Exoplanet Densities</i>   |
| 2021 | Isabel Colman (AMNH Postdoctoral researcher)<br>Project: <i>Measuring Stellar Rotation Periods from 200K TESS Time Series</i>                            |
| 2021 | Sarah Blunt (Caltech graduate student)<br>Project: <i>Modeling Stellar Radial Velocity Time Series with Gaussian Processes</i>                           |
| 2021 | Adina Feinstein (Univ. of Chicago graduate student)<br>Project: <i>Probabilistic Inference of Stellar and Planetary Parameters from TESS Time Series</i> |
| 2021 | Karl Jaehnig (Vanderbilt Univ. graduate student)<br>Project: <i>Modeling Stellar Binary Time Series with Gaussian Processes</i>                          |
| 2021 | Quang Tran (UT Austin graduate student)<br>Project: <i>Modeling Stellar Variability in Radial Velocities using Quasi-Periodic Gaussian Processes</i>     |
| 2021 | Pa Chia Thao (UNC Chapel Hill graduate student)<br>Project: <i>Time Series Forecasting with Recurrent Neural Networks</i>                                |
| 2020 | Angeli Sandoval (CUNY Hunter College undergraduate)<br>Project: <i>The Influence of Age on Exoplanet Demographics</i>                                    |
| 2018 | Grant Regen (Stanford Univ. undergraduate)<br>Project: <i>Spectroscopic Confirmation of Astrometrically-Selected Brown Dwarfs in Moving Groups</i>       |
| 2016 | Brianna Thomas (Howard Univ. undergraduate)<br>Project: <i>Evolution of Debris Disks</i>                                                                 |
| 2014 | Lyra Cao (Caltech undergraduate)<br>Project: <i>Bayesian Age Estimation of Pre-Main Sequence Stars</i>                                                   |

## SERVICE & OUTREACH

|           |                                                                                       |
|-----------|---------------------------------------------------------------------------------------|
| 2022      | Exoplanet Science Lead for NASA Small Explorer Mission Concept (\$150M class mission) |
| 2020–2022 | Exoplanet Group Meeting Leader, CCA, Flatiron Institute                               |
| 2015–2022 | Referee for <i>ApJ</i> , <i>ApJL</i> , <i>AJ</i> , <i>A&amp;A</i> , <i>MNRAS</i>      |
| 2022      | NASA XRP Grant Review Panelist                                                        |
| 2021      | Hubble Space Telescope Grant Review Panelist (Cycle 29)                               |
| 2020      | NASA ADAP Grant Review Panelist                                                       |
| 2020      | NASA TESS Grant Review Panelist (Cycle 3)                                             |
| 2018      | Astro2020 Decadal Survey White Paper Discussion Leader, NASA JPL                      |
| 2018      | European Research Council Starting Grant (€1.5M) Review Panelist                      |
| 2017      | Astro-ph Discussion Leader, Caltech                                                   |
| 2015      | Local Organizing Committee, ExSoCal, Caltech                                          |
| 2011      | Guest Lecturer, Cleveland Elementary School, Pasadena, CA                             |
| 2010–2011 | Volunteer, The Planetary Society, Pasadena, CA                                        |
| 2008–2009 | Society of Physics Students, Vassar College                                           |
| 2006–2009 | Physics and Astronomy Majors Committee, Vassar College                                |
| 2005–2009 | Observatory Assistant, Vassar College Observatory                                     |

## MEDIA COVERAGE

|         |                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------|
| 02/2026 | Physics (APS), <a href="#">Exoplanet Observations Sharpen Picture of Planetary Formation</a> .                          |
| 01/2026 | Nature News & Views, <a href="#">Ultra-low-density exoplanets observed around a young star</a> .                        |
| 01/2026 | New Scientist, <a href="#">Super-low-density worlds reveal how common planetary systems form</a> .                      |
| 01/2026 | BBC Science Focus, <a href="#">Most planets in our galaxy are born 'bloated', new study suggests</a> .                  |
| 01/2024 | The New York Times, <a href="#">This Distant Planet Has a 350,000-Mile-Long Cometlike Tail</a> .                        |
| 06/2022 | Simons Foundation Annual Report (cover story), <a href="#">Worlds Away</a> .                                            |
| 05/2021 | Simons Foundation Press Release, <a href="#">Shrinking Planets Could Explain Mystery of Universe's Missing Worlds</a> . |
| 11/2019 | Nature Astronomy Research Highlight, <a href="#">A Kepler multiplanet system precursor</a> .                            |
| 11/2019 | AAS Nova Research Highlight, <a href="#">Lessons from a Quartet of Newborn Planets</a> .                                |
| 06/2016 | Caltech Press Release, <a href="#">Newborn Exoplanet Discovered Around Young Star</a> .                                 |
| 06/2016 | NASA Press Release, <a href="#">NASA's K2 Finds Newborn Exoplanet Around Young Star</a> .                               |
| 06/2016 | The Guardian, <a href="#">Discovery of 'baby' planets sheds light on planet and solar system formation</a> .            |
| 06/2016 | Los Angeles Times, <a href="#">Newly discovered 'baby' planets could unlock mysteries of planetary evolution</a> .      |
| 06/2016 | The Washington Post, <a href="#">These brand new baby exoplanets could help us understand where worlds come from</a> .  |

## REFERENCES

References available upon request.

## PUBLICATIONS

14 lead author papers (979 citations), 60 total papers (3164 citations).

h-index: 32, g-index: 56, i10-index: 51. Updated June 8, 2026. Reverse chronological order.

## LEAD AUTHOR

1. **David, T. J.**, Angus, R., Curtis, J. L, and 5 colleagues, 2022, “Further Evidence of Modified Spin-down in Sun-like Stars: Pileups in the Temperature-Period Distribution,” *The Astrophysical Journal*, 933, 114. [\[ADS\]](#).
2. **David, T. J.**, Contardo, G., Sandoval, A., and 7 colleagues, 2021, “Evolution of the Exoplanet Size Distribution: Forming Large Super-Earths Over Billions of Years,” accepted to *The Astronomical Journal*. [\[ADS\]](#).
3. **David, T. J.**, Petigura, E. A., Luger, R., and 4 colleagues, 2019, “Four newborn planets transiting the young solar analog V1298 Tau,” *The Astrophysical Journal Letters*, 885, L12. [\[ADS\]](#).

4. **David, T. J.**, Cody, A. M., Hedges, C. L., and 14 colleagues, 2019, “A warm Jupiter-sized planet transiting the pre-main sequence star V1298 Tau,” *The Astronomical Journal*, 158, 79. [ADS].
5. **David, T. J.**, Hillenbrand, L. A., Gillen, E., and 4 colleagues, 2019, “Age Determination in Upper Scorpius with Eclipsing Binaries,” *The Astrophysical Journal*, 872, 161. [ADS].
6. **David, T. J.**, Mamajek, E. E., Vanderburg, A., and 19 colleagues, 2018, “Discovery of a Transiting Adolescent Sub-Neptune Exoplanet with K2,” *The Astronomical Journal*, 156, 302. [ADS].
7. **David, T. J.**, Crossfield, I. J. M., Benneke, B., and 14 colleagues, 2018, “Three Small Planets Transiting the Bright Young Field Star K2-233,” *The Astronomical Journal*, 155, 222. [ADS].
8. **David, T. J.**, 2018, “On the Evolutionary Pathways of Stars and Extrasolar Planets,” Ph.D. Thesis, *California Institute of Technology*. [ADS].
9. **David, T. J.**, Petigura, E. A., Hillenbrand, L. A., and 14 colleagues, 2017, “A Transient Transit Signature Associated with the Young Star RIK-210,” *The Astrophysical Journal*, 835, 168. [ADS].
10. **David, T. J.**, Hillenbrand, L. A., Petigura, E. A., and 10 colleagues, 2016, “A Neptune-sized transiting planet closely orbiting a 5-10-million-year-old star,” *Nature*, 534, 658. [ADS].
11. **David, T. J.**, Conroy, K. E., Hillenbrand, L. A., and 7 colleagues, 2016, “New Pleiades Eclipsing Binaries and a Hyades Transiting System Identified by K2,” *The Astronomical Journal*, 151, 112. [ADS].
12. **David, T. J.**, Hillenbrand, L. A., Cody, A. M., and 5 colleagues, 2016, “K2 Discovery of Young Eclipsing Binaries in Upper Scorpius: Direct Mass and Radius Determinations for the Lowest Mass Stars and Initial Characterization of an Eclipsing Brown Dwarf Binary,” *The Astrophysical Journal*, 816, 21. [ADS].
13. **David, T. J.**, Stauffer, J. R., Hillenbrand, L. A., and 16 colleagues, 2015, “HII 2407: An Eclipsing Binary Revealed By K2 Observations of the Pleiades,” *The Astrophysical Journal*, 814, 62. [ADS].
14. **David, T. J.**, and Hillenbrand, L. A., 2015, “The Ages of Early-type Stars: Strömgren Photometric Methods Calibrated, Validated, Tested, and Applied to Hosts and Prospective Hosts of Directly Imaged Exoplanets,” *The Astrophysical Journal*, 804, 146. [ADS].

#### ANY AUTHOR

1. Livingston, J. H., Petigura, E. A., **David, T. J.**, and 39 colleagues, 2026, “A young progenitor for the most common planetary systems in the Galaxy,” *Nature*, 649, 310–314.
2. Chance, Q., Foreman-Mackey, D., Ballard, S., Casey, A. R., **David, T. J.**, and 1 colleague, 2025, “paired: A Statistical Framework for Detecting Stellar Binarity with Gaia RVs. I. Sensitivity to Unresolved Binaries,” *The Astrophysical Journal*, 992, 131. [ADS].
3. Barat, S., Désert, J. M., and 20 colleagues, 2025, “A Metal-poor Atmosphere with a Hot Interior for a Young Sub-Neptune Progenitor: JWST/NIRSpec Transmission Spectrum of V1298 Tau b,” *The Astronomical Journal*, 170, 165. [ADS].
4. Barat, S., Désert, J. M., Goyal, J. M., and 13 colleagues, 2024, “First comparative exoplanetology within a transiting multi-planet system: Comparing the atmospheres of V1298 Tau b and c,” *Astronomy & Astrophysics*, 692, A198. [ADS].
5. Barat, S., Désert, J. M., Vazan, A., and 12 colleagues, 2024, “The metal-poor atmosphere of a potential Sub-Neptune progenitor,” *Nature Astronomy*, 8, 899. [ADS].
6. Colman, I. L., Angus, R., **David, T. J.**, and 3 colleagues, 2024, “Methods for the Detection of Stellar Rotation Periods in Individual TESS Sectors and Results from the Prime Mission,” *The Astronomical Journal*, 167, 189. [ADS].
7. Tyler, D., Petigura, E. A., Oklopčić, A., and **David, T. J.**, 2024, “WASP-69b’s Escaping Envelope Is Confined to a Tail Extending at Least  $7 R_p$ ,” *The Astrophysical Journal*, 960, 123. [ADS].
8. Blunt, S., Carvalho, A., **David, T. J.**, and 23 colleagues, 2023, “Overfitting Affects the Reliability of Radial Velocity Mass Estimates of the V1298 Tau Planets,” *The Astronomical Journal*, 166, 62. [ADS].

9. Jaehnig, K., Price-Whelan, A. M., Foreman-Mackey, D. and 5 colleagues, “Spectroscopic eclipsing binary orbital parameter estimation with TESS and APOGEE,” submitted to AAS Journals.
10. Chance, Q., Foreman-Mackey, D., Ballard, S., and 3 colleagues, 2022, “paired: A Statistical Framework for Determining Stellar Binarity with Gaia RVs. I. Planet Hosting Binaries,” submitted to *The Astrophysical Journal*. [\[ADS\]](#).
11. Thao, P. C., Mann, A. W., Gao, P., and 10 colleagues, 2023, “Hazy with a Chance of Star Spots: Constraining the Atmosphere of Young Planet K2-33b,” *The Astronomical Journal*, 165, 23. [\[ADS\]](#).
12. Lu, Y. L., Curtis, J. L., Angus, R., and 2 colleagues, 2022, “Bridging the Gap-The Disappearance of the Intermediate Period Gap for Fully Convective Stars, Uncovered by New ZTF Rotation Periods,” *The Astronomical Journal*, 164, 251. [\[ADS\]](#).
13. **David, T. J.**, Angus, R., Curtis, J. L., and 5 colleagues, 2022, “Further Evidence of Modified Spin-down in Sun-like Stars: Pileups in the Temperature-Period Distribution,” *The Astrophysical Journal*, 933, 114. [\[ADS\]](#).
14. Johnson, M. C., **David, T. J.**, Petigura, E. A., and 22 colleagues, 2022, “An Aligned Orbit for the Young Planet V1298 Tau b,” *The Astronomical Journal*, 163, 247. [\[ADS\]](#).
15. Hedges, C., Hughes, A., Zhou, G., and 49 colleagues, 2022, “Erratum: “TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up” (2021, AJ, 162, 54),” *The Astronomical Journal*, 163, 143. [\[ADS\]](#).
16. Feinstein, A. D., **David, T. J.**, Montet, B. T., and 3 colleagues, 2022, “V1298 Tau with TESS: Updated Ephemerides, Radii, and Period Constraints from a Second Transit of V1298 Tau e,” *The Astrophysical Journal Letters*, 925, L2. [\[ADS\]](#).
17. Hedges, C., Hughes, A., Zhou, G., and 49 colleagues, 2021, “Erratum: “TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up” (2021, AJ, 162, 54),” *The Astronomical Journal*, 162, 305. [\[ADS\]](#).
18. Vissapragada, S., Stefánsson, G., Greklek-McKeon, M., and 28 colleagues, 2021, “A Search for Planetary Metastable Helium Absorption in the V1298 Tau System,” *The Astronomical Journal*, 162, 222. [\[ADS\]](#).
19. Feinstein, A., Montet, B., Johnson, M., and 5 colleagues, 2021, “H- $\alpha$  and Ca II Infrared Triplet Variations During a Transit of the 23 Myr Planet V1298 Tau c,” *The Astronomical Journal*, 162, 213. [\[ADS\]](#).
20. Rebull, L., Stauffer, J. R., Cody, A. M., and 4 colleagues, 2021, “Erratum: “Rotation of Low-mass Stars in Taurus with K2” (2020, AJ, 159, 273),” *The Astronomical Journal*, 162, 172. [\[ADS\]](#).
21. Hedges, C., Hughes, A., Zhou, G., **David, T. J.**, and 43 colleagues, 2021, “TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up,” *The Astronomical Journal*, 162, 54. [\[ADS\]](#).
22. Foreman-Mackey, D., Luger, R., Agol, E., and 13 colleagues, 2021, “exoplanet: Gradient-based probabilistic inference for exoplanet data & other astronomical time series,” *The Journal of Open Source Software*, 6, 3285. [\[ADS\]](#).
23. **David, T. J.**, Contardo, G., Sandoval, A., and 7 colleagues, 2020, “Evolution of the Exoplanet Size Distribution: Forming Large Super-Earths Over Billions of Years,” *The Astronomical Journal*, 161, 265. [\[ADS\]](#).
24. Sandoval, A., Contardo, G., and **David, T. J.**, 2020, “The Influence of Age on the Relative Frequency of Super-Earths and Sub-Neptunes,” *The Astrophysical Journal*, 911, 117. [\[ADS\]](#).
25. Lu, Y., Angus, R., Curtis, J. L., and 2 colleagues, 2021, “Gyro-Kinematic Ages for around 30,000 Kepler Stars,” *The Astronomical Journal*, 161, 189. [\[ADS\]](#).
26. Matthews, E. C., Hinkley, S., Stapelfeldt, K., and 8 colleagues, 2020, “Three new late-type stellar companions to very dusty WISE debris disks identified with VLT/SPHERE imaging,” *The Astronomical Journal*, 161, 78. [\[ADS\]](#).
27. van Dam, D. M., Kenworthy, M. A., **David, T. J.**, and 21 colleagues, 2020, “An Asymmetric Eclipse Seen toward the Pre-main-sequence Binary System V928 Tau,” *The Astronomical Journal*, 160, 285. [\[ADS\]](#).
28. Gagné, J., **David, T. J.**, Mamajek, E. E., and 3 colleagues, 2020, “The  $\mu$  Tau Association: A 60 Myr Old Coeval Group at 150 pc from the Sun,” *The Astrophysical Journal*, 903, 96. [\[ADS\]](#).



29. Stauffer, J. R., Barrado, D., **David, T. J.** and 7 colleagues, 2020, “Pleiades or Not? Resolving the Status of the Lithium-rich M Dwarfs HHJ 339 and HHJ 430,” *The Astronomical Journal*, 160, 30. [ADS].
30. Rebull, L. M., Stauffer, J. R., Cody, A. M. and 4 colleagues, 2020, “Rotation of Low-mass Stars in Taurus with K2,” *The Astronomical Journal*, 159, 273. [ADS].
31. **David, T. J.**, Petigura, E. A., Luger, R., and 4 colleagues, 2019, “Four newborn planets transiting the young solar analog V1298 Tau,” *The Astrophysical Journal Letters*, 885, L12. [ADS].
32. Hippke, M., **David, T. J.**, Mulders, G. D., and Heller, R., 2019, “Wotan: Comprehensive time-series de-trending in Python,” *The Astronomical Journal*, 158, 143. [ADS].
33. **David, T. J.**, Cody, A. M., Hedges, C. L., and 14 colleagues, 2019, “A warm Jupiter-sized planet transiting the pre-main sequence star V1298 Tau,” *The Astronomical Journal*, 158, 79. [ADS].
34. Beichman, C. A., Hirano, **David, T. J.**, and 9 colleagues, 2019, “A Mass Limit for the Young Transiting Planet V1298 Tau b,” *Research Notes of the AAS*, 3, 6. [IOP].
35. **David, T. J.**, Hillenbrand, L. A., Gillen, E., and 4 colleagues, 2019, “Age Determination in Upper Scorpius with Eclipsing Binaries,” *The Astrophysical Journal*, 872, 161. [ADS].
36. Mellon, S. N., Mamajek, E. E., Zwintz, K., and 11 colleagues, 2019, “Discovery of  $\delta$  Scuti Pulsations in the Young Hybrid Debris Disk Star HD 156623,” *The Astrophysical Journal*, 870, 36. [ADS].
37. **David, T. J.**, Mamajek, E. E., Vanderburg, A., and 19 colleagues, 2018, “Discovery of a Transiting Adolescent Sub-Neptune Exoplanet with K2,” *The Astronomical Journal*, 156, 302. [ADS].
38. Stauffer, J. R., Rebull, L. M., Cody, A. M., and 5 colleagues, 2018, “The Rotational Evolution of Young, Binary M Dwarfs,” *The Astronomical Journal*, 156, 275. [ADS].
39. Crossfield, I. J. M., Guerrero, N., **David, T. J.**, and 38 colleagues, 2018, “A TESS Dress Rehearsal: Planetary Candidates and Variables from K2 Campaign 17,” *The Astrophysical Journal Supplement*, 239, 5. [ADS].
40. Peterson, M. S., Benneke, B., **David, T. J.**, and 21 colleagues, 2018, “A  $2 R_{\oplus}$  Planet Orbiting the Bright Nearby K Dwarf Wolf 503,” *The Astronomical Journal*, 156, 188. [ADS].
41. Matthews, E., Hinkley, S., Vigan, A., and 10 colleagues, 2018, “Constraining the presence of giant planets in two-belt debris disc systems with VLT/SPHERE direct imaging and dynamical arguments,” *Monthly Notices of the Royal Astronomical Society*, 480, 2757. [ADS].
42. Wang, J., **David, T. J.**, Hillenbrand, L. A., and 3 colleagues, 2018, “EPIC 203868608: A Low-mass Quadruple Star System in the Upper Scorpius OB Association,” *The Astrophysical Journal*, 865, 141. [ADS].
43. **David, T. J.**, Crossfield, I. J. M., Benneke, B., and 14 colleagues, 2018, “Three Small Planets Transiting the Bright Young Field Star K2-233,” *The Astronomical Journal*, 155, 222. [ADS].
44. Rebull, L. M., Stauffer, J. R., Cody, A. M., and 3 colleagues, 2018, “Rotation of Low-mass Stars in Upper Scorpius and  $\rho$  Ophiuchus with K2,” *The Astronomical Journal*, 155, 196. [ADS].
45. Stauffer, J. R., Rebull, L. M., **David, T. J.**, and 8 colleagues, 2018, “More Rapidly Rotating PMS M Dwarfs with Light Curves Suggestive of Orbiting Clouds of Material,” *The Astronomical Journal*, 155, 63. [ADS].
46. **David, T. J.**, 2018, “On the Evolutionary Pathways of Stars and Extrasolar Planets,” Ph.D. Thesis, *California Institute of Technology*. [ADS].
47. Ciardi, D. R., Crossfield, I. J. M., Feinstein, A. D., and 14 colleagues, 2018, “K2-136: A Binary System in the Hyades Cluster Hosting a Neptune-sized Planet,” *The Astronomical Journal*, 155, 10. [ADS].
48. Gillen, E., Hillenbrand, L. A., **David, T. J.**, and 5 colleagues, 2017 “New Low-mass Eclipsing Binary Systems in Praesepe Discovered by K2,” *The Astrophysical Journal*, 849, 11. [ADS].
49. Pepper, J., Gillen, E., Parviainen, H., and 11 colleagues, 2017, “A Low-mass Exoplanet Candidate Detected by K2 Transiting the Praesepe M Dwarf JS 183,” *The Astronomical Journal*, 153, 177. [ADS].
50. Stauffer, J. R., Collier Cameron, A., Jardine, M., and 8 colleagues, 2017, “Orbiting Clouds of Material at the Keplerian Co-rotation Radius of Rapidly Rotating Low-mass WTTs in Upper Sco,” *The Astronomical Journal*, 153, 152. [ADS].



51. Cody, A. M., Hillenbrand, L. A., **David, T. J.**, and 3 colleagues, 2017, “A Continuum of Accretion Burst Behavior in Young Stars Observed by K2,” *The Astrophysical Journal*, 836, 41. [\[ADS\]](#).
52. **David, T. J.**, Petigura, E. A., Hillenbrand, L. A., and 14 colleagues, 2017, “A Transient Transit Signature Associated with the Young Star RIK-210,” *The Astrophysical Journal*, 835, 168. [\[ADS\]](#).
53. Obermeier, C., Henning, T., Schlieder, J. E., and 21 colleagues, 2016, “K2 Discovers a Busy Bee: An Unusual Transiting Neptune Found in the Beehive Cluster,” *The Astronomical Journal*, 152, 223. [\[ADS\]](#).
54. Stauffer, J. R., Rebull, L. M., Bouvier, J., and 15 colleagues, 2016, “Rotation in the Pleiades with K2. III. Speculations on Origins and Evolution,” *The Astronomical Journal*, 152, 115. [\[ADS\]](#).
55. **David, T. J.**, Hillenbrand, L. A., Petigura, E. A., and 10 colleagues, 2016, “A Neptune-sized transiting planet closely orbiting a 5-10-million-year-old star,” *Nature*, 534, 658. [\[ADS\]](#).
56. **David, T. J.**, Conroy, K. E., Hillenbrand, L. A., and 7 colleagues, 2016, “New Pleiades Eclipsing Binaries and a Hyades Transiting System Identified by K2,” *The Astronomical Journal*, 151, 112. [\[ADS\]](#).
57. **David, T. J.**, Hillenbrand, L. A., Cody, A. M., and 5 colleagues, 2016, “K2 Discovery of Young Eclipsing Binaries in Upper Scorpius: Direct Mass and Radius Determinations for the Lowest Mass Stars and Initial Characterization of an Eclipsing Brown Dwarf Binary,” *The Astrophysical Journal*, 816, 21. [\[ADS\]](#).
58. **David, T. J.**, Stauffer, J. R., Hillenbrand, L. A., and 16 colleagues, 2015, “HII 2407: An Eclipsing Binary Revealed By K2 Observations of the Pleiades,” *The Astrophysical Journal*, 814, 62. [\[ADS\]](#).
59. Mawet, D., **David, T. J.**, Bottom, M., and 7 colleagues, 2015, “Discovery of a Low-mass Companion Around HR 3549,” *The Astrophysical Journal*, 811, 10. [\[ADS\]](#).
60. **David, T. J.**, and Hillenbrand, L. A., 2015, “The Ages of Early-type Stars: Strömgren Photometric Methods Calibrated, Validated, Tested, and Applied to Hosts and Prospective Hosts of Directly Imaged Exoplanets,” *The Astrophysical Journal*, 804, 146. [\[ADS\]](#).
61. Hinkley, S., Pueyo, L., Faherty, J. K., and 29 colleagues, 2013, “The  $\kappa$  Andromedae System: New Constraints on the Companion Mass, System Age, and Further Multiplicity,” *The Astrophysical Journal*, 779, 153. [\[ADS\]](#).
62. Sheffield, A. A., Majewski, S. R., Johnston, K. V., and 10 colleagues, 2012, “Identifying Contributions to the Stellar Halo from Accreted, Kicked-out, and In Situ Populations,” *The Astrophysical Journal*, 761, 161. [\[ADS\]](#).